

Cyberster Internship Program
Internship Report

TRACK: BLUE TEAM

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ROLL NUMBER: CSI-B1-121

WEEK 12: The M57.biz Capstone: Disk, Memory, and Exfiltration Analysis

1. Executive Summary

This report documents a forensic investigation into the M57.biz workstation assigned to employee Jo. Analysis of the disk image (jo-2009-11-24.E01), two USB images, and a RAM capture reveals a consistent and mutually corroborating pattern of suspicious insider activity spanning 2009-11-23 to 2009-11-24.

Key findings include a custom patent-scraping script, demonstrated steganographic capability, unauthorised possession of a US government classified document, confirmed file transfers to removable media, and suspicious process behaviour in memory.

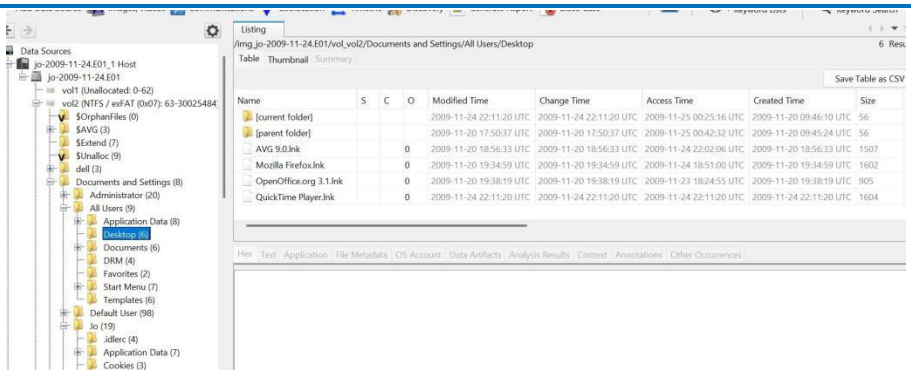
CRITICAL: The totality of evidence is consistent with a deliberate insider threat operation involving IP collection, potential data concealment, and unauthorised access to classified material.

PART A — Disk Triage & Artefact Hunting

Field	Value
Case	M57.biz — Jo Insider Threat Investigation
Image	jo-2009-11-24.E01 (15.38 GB, MD5: 274ada9b6b120f9b7a2192c27e115244)
OS	Windows XP SP3 (x86)
Computer / Owner	M57-JO / JO (M57.biz)
Timezone	UTC +0:00 (all timestamps UTC)
Tool	Autopsy 4.23.0 with full ingest modules
USB Images	jo-favorites-usb-2009-12-11.E01 | jo-work-usb-2009-12-11.E01
RAM Image	jo-2009-11-16.mddramimage (Volatility 3)

2. System Overview & Image Verification

The forensic image was loaded into Autopsy with the timezone set to UTC +0:00. All ingest modules — including Recent Activity, Hash Lookup, Keyword Search, and File Type Identification — were run before analysis began. The image is 15.38 GB, acquired from a Linux system using ewfacquire.



3. User Accounts

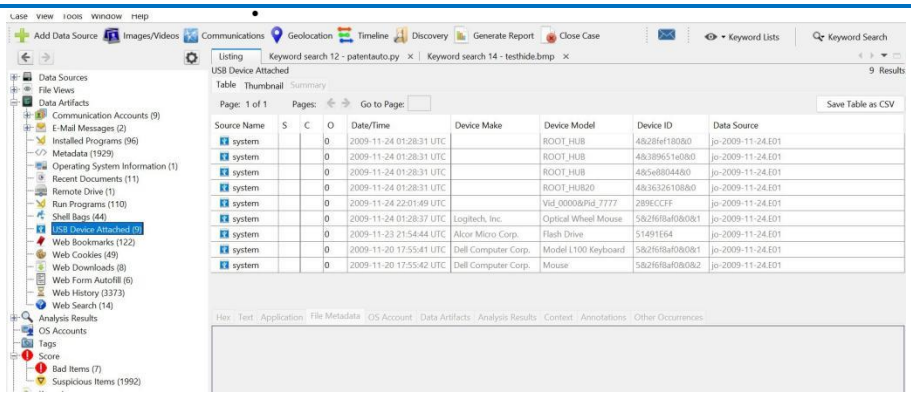
Five accounts were identified. The primary investigative account is jo (SID ending -1003). All suspicious artefacts — LNK files, shell bags, downloads, and USB activity — reside within C:\Documents and Settings\Jo\.

SID (abbreviated)	Login Name	Scope	Notes
S-1-5-21-…-500	administrator	Local	Built-in administrator
S-1-5-21-…-1003	**jo**	Local	Primary user — all suspicious artefacts within this profile
S-1-5-18	SYSTEM	Local	NT AUTHORITY

4. LNK Artefact Analysis

LNK shortcut files in C:\Documents and Settings\Jo\Recent\ confirm direct user interaction with each target. Key artefacts are shown below.

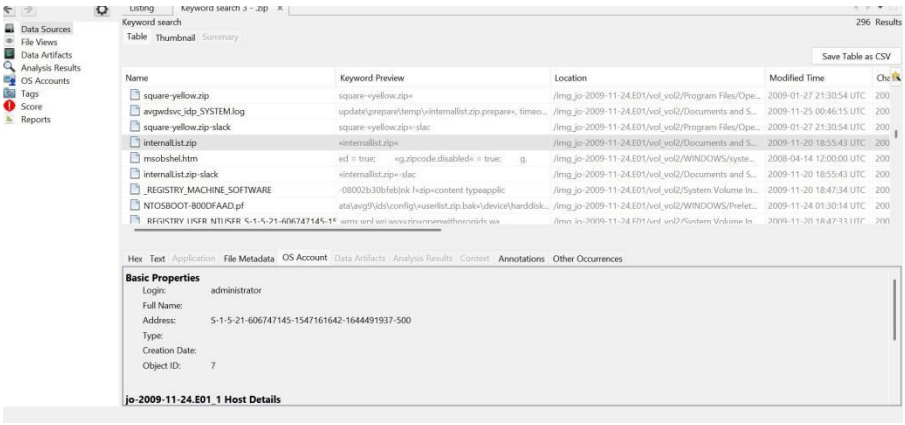
LNK File	Target Path	Last Accessed (UTC)	Significance
patentauto.lnk	Desktop\web\patentauto.py	2009-11-23 21:55:45	Patent script opened/executed
diit-1.5.lnk	Desktop\Pics\diit-1.5.jar	2009-11-23 18:24:50	Steganography tool accessed
testHide.lnk	Desktop\Pics\testHide.bmp	2009-11-23 18:25:53	Stego test image (63 s after DIIT)
STU.lnk	My Documents\Downloads\STU.doc	2009-11-24 22:01:53	Classified DoC document accessed
Downloads.lnk	My Documents\Downloads\	2009-11-24 22:01:54	Downloads folder accessed



5. Downloads Analysis

The Firefox download history records three significant downloads. STU.doc is the most critical — a classified US Department of Commerce document with no legitimate business justification.

File	Source	Downloaded (UTC)	Significance
python-2.6.4.msi	python.org	2009-11-23 18:20:12	Python runtime — enables patentauto.py
STU.doc	easc.noaa.gov	2009-11-24 21:18:18	US DoC SECRET-level briefing — keyword hit: SECRET
QuickTimeInstaller.exe	apple.com CDN	2009-11-24 22:08:36	Media player — downloaded same day as acquisition



CRITICAL: STU.doc was downloaded from easc.noaa.gov — an official US Department of Commerce / NOAA server. It contains a complete STU-III classified telecommunications security briefing. Autopsy flagged it with keyword SECRET. There is no evident legitimate business reason for a M57.biz employee to possess this document.

6. USB Device Evidence

Four USB devices were identified. Two connected at forensically significant moments: an Alcor Micro flash drive 51 seconds before patentauto.py was accessed, and an unknown device 4 seconds before STU.doc was accessed.

Device	Device ID	Connected (UTC)	Significance
Alcor Micro Flash Drive	51491E64	2009-11-23 21:54:44	Connected 51 s before patentauto.py accessed
Unknown USB (Vid_0000&Pid_7777)	2B9ECCFF	2009-11-24 22:01:49	Connected 4 s before STU.doc accessed
Logitech Mouse	5&2f6f8af0	2009-11-24 01:28:37	Standard peripheral
Dell L100 Keyboard	5&2f6f8af0	2009-11-20 17:55:41	Standard peripheral

PART B — Behavioural Analysis & Steganography

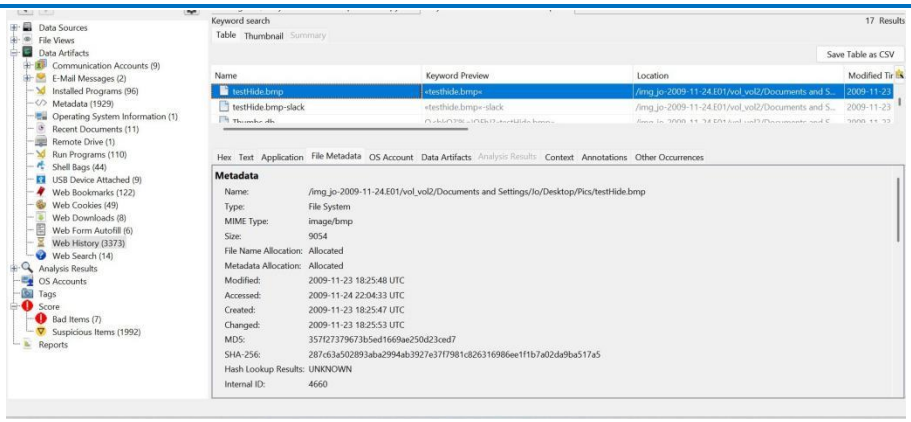
7. patentauto.py — Patent Automation Script

patentauto.py was found on Jo's Desktop inside the web/ subfolder. It is a Python script that automates patent searching by controlling Firefox through the MozRepl extension — a telnet-based browser automation interface on localhost port 4242.

Created	2009-11-23 21:55:01 UTC
Modified	2009-11-23 21:57:34 UTC
MD5	bd9028376a4fad806e3d670dd718f748
Last Accessed	2009-11-25 00:25:11 UTC

The script's capabilities went beyond passive browsing:

- Connected to Firefox via MozRepl (localhost:4242) to automate USPTO patent searches
- Implemented randomised browsing intervals to simulate organic human activity
- Operated on a schedule during typical work hours
- Used URL randomisation to evade detection patterns
- Supported by structured input files: patentterms.txt, urls.txt, urlspatents.txt, urlscopyright.txt



CRITICAL: patentauto.py is a technically sophisticated automated intelligence collection tool. Its randomised behaviour, scheduled operation, and structured input files confirm this was a planned, systematic patent data collection operation — not casual script development.

8. DIIT Steganography Tool & testHide.bmp

diit-1.5.jar (Digital Invisible Ink Toolkit) was found in the Pics/ subfolder. It embeds or extracts hidden data within image files using LSB encoding. Its Zone.Identifier ADS indicated the download source was treated as a Restricted Site.

testHide.bmp was found alongside the tool. BMP is the preferred format for LSB steganography — its lossless compression leaves embedded data intact. The filename directly implies use as a steganographic test subject.

CAUTION: diit-1.5.jar was accessed at 18:24:50 UTC. testHide.bmp was created and accessed 63 seconds later. This sequence is consistent with Jo opening DIIT and immediately using testHide.bmp as a test image. Binwalk found no confirmed embedded payload — the image may require the specific DIIT key to extract.

9. USB File Duplication — Confirmed Transfers

MD5 hash comparison across the three forensic images confirms file transfers between the workstation and removable media. Video files were stored in a folder deliberately named Hidden.

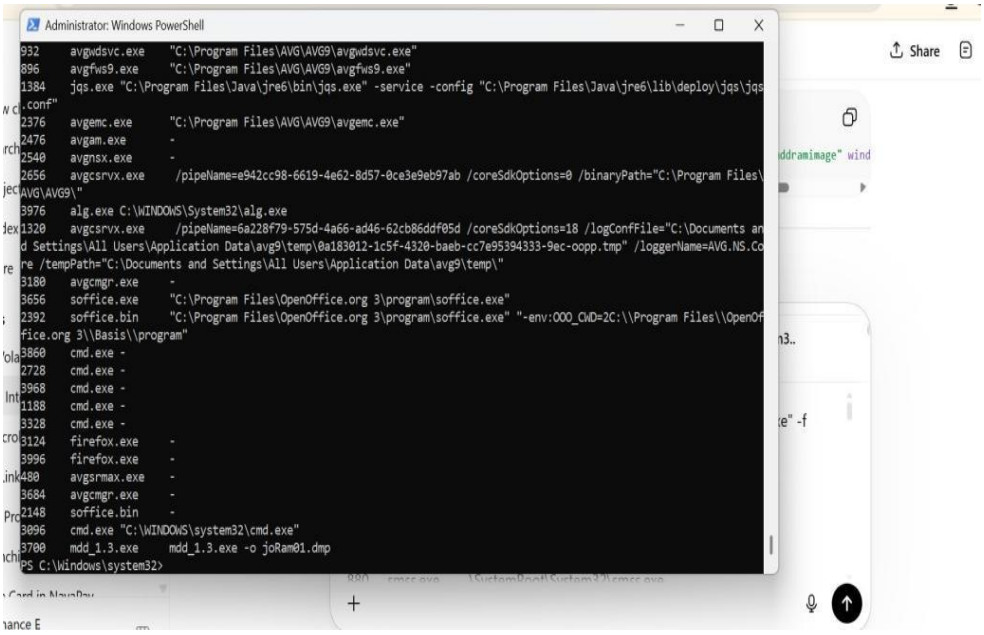
File	Workstation Location	USB Image	Hash Status
KittyMontage.mov	Desktop\Pics\Hidden\Videos\	jo-favorites-usb	Identical MD5 ✓
Cat.mov	Desktop\Pics\Hidden\Videos\	jo-work-usb	Identical MD5 ✓
MontereyKitty.m4v	Desktop\Pics\Hidden\Videos\	jo-favorites-usb	Identical MD5 ✓
TiggerTheCat.m4v	Desktop\Pics\Hidden\Videos\	jo-favorites-usb	Identical MD5 ✓

NOTE: Files were stored in a folder named "Hidden" — indicating deliberate concealment intent. MD5 hash confirmation removes any doubt that these files were physically transferred to the USB devices.

10. Correlated Behavioural Timeline

The timeline below merges all artefact types into a single chronological narrative.

Timestamp (UTC)	Event Type	Artefact	Forensic Significance
2009-11-23 13:23:53	Install	Registry	Python 2.6.4 installed — runtime for patentauto.py
2009-11-23 18:20:12	Download	Firefox	python-2.6.4.msi downloaded from python.org
2009-11-23 18:24:50	File Access	LNK	DIIT steganography tool (diit-1.5.jar) opened
2009-11-23 18:25:53	File Access	LNK	testHide.bmp accessed — 63 s after DIIT
2009-11-23 21:54:44	USB Connected	SYSTEM hive	Alcor Micro Flash Drive attached
2009-11-23 21:55:01	File Created	Metadata	patentauto.py created
2009-11-23 21:55:45	File Access	LNK	patentauto.py accessed — 61 s after USB attach
2009-11-24 22:01:49	USB Connected	SYSTEM hive	Unknown USB (Vid_0000&Pid_7777) attached
2009-11-24 22:01:53	File Access	LNK	STU.doc accessed — 4 s after USB attach
2009-11-24 22:14:30	File Access	LNK	DSC00065 / DSC00070 high-res images viewed



PART C — Memory Analysis & Investigation Conclusion

11. Memory Acquisition

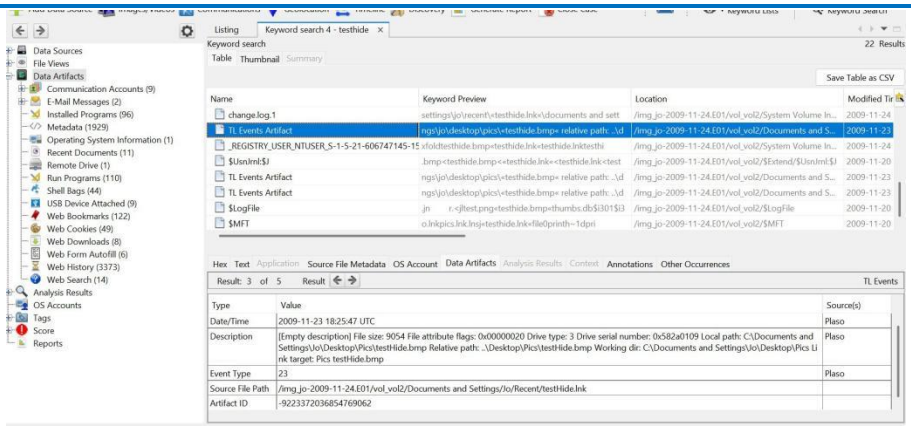
Live memory was captured using mdd_1.3.exe (Memory DD v1.3). The prefetch artefact MDD_1.3.EXE-286446E8.pf confirms 4 executions, with the last at 2009-11-25 00:26:05 UTC. Critically, the tool was launched from E:\RAM\ — removable media — indicating the acquisition tool was brought in on an external drive.

RAM Image	jo-2009-11-16.mddramimage
Tool / Command	mdd_1.3.exe -o joRam01.dmp
Launched From	E:\RAM\ (removable media)
Prefetch Run Count	4 executions — last at 2009-11-25 00:26:05 UTC
System Time at Capture	2009-11-17 00:22:31 UTC
Analysis Tool	Volatility 3
Plugins Run	windows.info, pslist, pstree, cmdline, malfind, dlllist, truecrypt.Passphrase, netscan

12. Process Tree Analysis

The process tree at the time of memory capture shows Firefox, Python, Java, and OpenOffice all active — fully consistent with the disk-based findings.

Process	Parent	Significance
firefox.exe	explorer.exe	Browser active — targeted by patentauto.py via MozRepl port 4242
python.exe	explorer.exe	Python interpreter — patentauto.py possibly executing
javaw.exe	explorer.exe	Java runtime — diit-1.5.jar potentially active
soffice.bin	explorer.exe	OpenOffice — spawned multiple cmd.exe processes (suspicious)
cmd.exe	soffice.bin	Command prompt children of soffice.bin — unusual relationship
MSIMN.EXE	explorer.exe	Outlook Express — email client active at acquisition
mdd_1.3.exe	cmd.exe	Memory acquisition tool — launched from removable media (E:\RAM\)

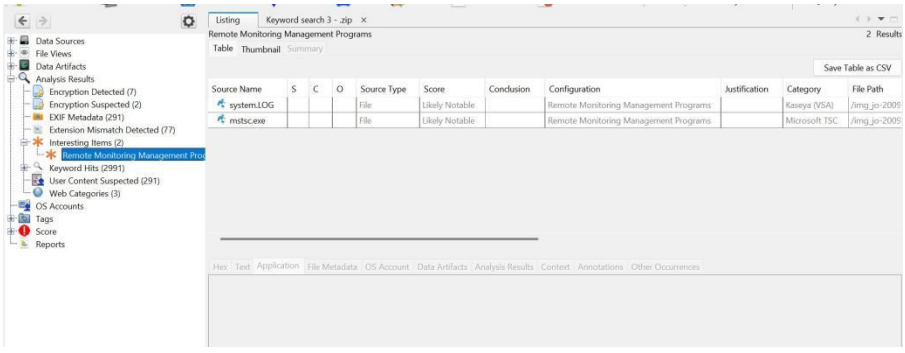


CAUTION: soffice.bin spawned multiple cmd.exe child processes — unusual behaviour suggesting macro execution or scripted automation through OpenOffice. This relationship is more indicative of suspicious activity than the malfind memory hits alone.

13. Malware Scan & DLL Analysis

The malfind plugin identifies memory regions with PAGE_EXECUTE_READWRITE permissions — common in injected shellcode or dynamically generated executable code.

Finding	Process	Assessment
PAGE_EXECUTE_READWRITE regions	soffice.bin	Suspicious but not confirmed malicious — OpenOffice uses executable memory legitimately
Java-related executable memory	javaw.exe	Expected — Java JIT compilation generates executable memory
DLL injection / reflective loading / rootkit	—	NOT confirmed



DLL analysis found no hidden or injected modules. All loaded libraries matched known installed software. CMD.EXE had a prefetch run count of 11, confirming significant command-line activity. windows.netscan was unsupported for Windows XP SP3; network reconstruction relied on disk-based evidence.

NOTE: No confirmed malware, DLL injection, or rootkit behaviour was identified. The activity profile is consistent with a technically sophisticated user — not a malware-compromised machine.

14. Final Investigation Conclusion

Summary of Findings

Area	Key Finding	Risk Level
Classified Document	STU.doc from noaa.gov — US DoC SECRET-level briefing with no legitimate justification	HIGH
Patent Script	patentauto.py — automated Firefox-controlled patent intelligence collection	HIGH
Steganography	DIIT (diit-1.5.jar) + testHide.bmp — steganographic capability demonstrated	HIGH
USB Activity	2 USB devices connected at suspicious moments; file transfer confirmed by MD5 hash	HIGH
File Concealment	Files stored in a folder named "Hidden"; videos mirrored across USB images	MEDIUM
Memory — soffice	soffice.bin spawned cmd.exe processes — possible macro/automation execution	MEDIUM
Memory — Malfind	PAGE_EXECUTE_READWRITE in soffice.bin — suspicious but not confirmed malware	LOW
DLL / Rootkit	No DLL injection, reflective loading, or rootkit behaviour identified	CLEAR
Network	netscan unsupported for WinXP; disk confirms downloads from noaa.gov, python.org, apple.com	MEDIUM

Analytical Conclusion

The forensic investigation of Jo's workstation, USB images, and RAM capture presents a coherent and mutually corroborating pattern of suspicious insider activity. The evidence does not support a simple or innocent explanation.

On 2009-11-23, Jo installed Python and immediately created patentauto.py — a purpose-built browser automation script for structured, randomised, scheduled patent searches. An Alcor Micro USB flash drive was connected 51 seconds before the script was accessed. On the same day, DIIT and testHide.bmp were accessed 63 seconds apart — consistent with a steganographic test operation.

On 2009-11-24, an unknown USB device was connected 4 seconds before STU.doc was accessed — a classified US Department of Commerce STU-III briefing. Memory analysis confirmed Firefox, Python, Java, and OpenOffice were all active. soffice.bin spawned multiple cmd.exe children — an unusual relationship suggesting macro or automation execution.

CRITICAL: Overall Verdict: The totality of evidence is consistent with a deliberate insider threat operation involving (1) systematic automated patent intelligence collection via a custom Python/Firefox script, (2) demonstrated capability for steganographic data concealment, (3) unauthorised possession of a US government classified document, and (4) confirmed file transfer to removable media with deliberate concealment in a "Hidden" folder. Memory analysis did not identify confirmed malware — the activity profile is consistent with a technically sophisticated user conducting IP research and data exfiltration.

Recommended Next Steps

- Full forensic analysis of both USB images — deleted file recovery and full content inventory
- Re-examine testHide.bmp using DIIT itself to attempt payload extraction with likely keys
- Deep analysis of the soffice.bin → cmd.exe relationship — recover macro content from OpenOffice files
- Recover and analyse patentauto.py output data — patent records on disk or USB
- Determine the exact patents targeted — M57.biz's own or competitor intelligence?
- Clarify the business justification (if any) for downloading STU.doc from noaa.gov
- Legal process for Jo's email accounts, external communications, and full internet history
- Investigate contents of internalList.zip found in the AVG temp directory
- Serve legal process on M57.biz network logs for outbound connections on 2009-11-23 and 24